



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Name: Muhammad Musawar Ali Shah

Roll No. 24P-0619

Dept: BS-CS

Section: BCS-3C

Subject: Data Structures

Submitted To:

Sir Saad Khan

Lab 4:

Question 1:

```
#include<iostream>
#include<string> // Added for string
using namespace std;

class Student{
private:
    int ID;
    string name;
    float gpa;
public:
    Student(int id = 0, string n = "", float g = 0.0) : ID(id), name(n), gpa(g) {}

    Student(const Student &s) {
        ID = s.ID;
        name = s.name;
        gpa = s.gpa;
    }

    void setID(int id) {
        ID = id;
    }

    void setName(string n) {
        name = n;
    }

    void setGpa(float Gpa) {
        if(Gpa >= 0.0 && Gpa <= 4.0) {
            gpa = Gpa;
        } else {
            cout << "Invalid GPA. GPA must be between 0.0 & 4.0" << endl;
        }
    }

    int getID() const { return ID;}

    string getName() const {return name;}

    float getGpa() const { return gpa; }
};

class Node{
```

```

public:
    Student data;
    Node *next;
    Node(Student val) : data(val), next(NULL) {}
};

class LinkedList{
private:
    Node *head;
public:
    LinkedList() : head(NULL) {}

    void addStudent(const Student& s) {
        Node *newNode = new Node(s);
        if(head == NULL) {
            head = newNode;
            newNode->next = head;
        } else {
            Node* temp = head;
            while (temp->next != head) {
                temp = temp->next;
            }
            temp->next = newNode;
            newNode->next = head;
        }
        cout << "Student added successfully." << endl;
    }

    void removeStudent(int studentID) {
        if(head == NULL) {
            cout << "Student not found." << endl;
            return;
        }
        Node* cN = head;
        Node* pN = NULL;

        if(cN->data.getID() == studentID) {
            if(cN->next == head) {
                delete head;
                head = NULL;
                cout << "Student removed successfully." << endl;
                return;
            }
        }

        Node* temp = head;
        while(temp->next != head) {
            temp = temp->next;
        }
    }

```

```

        head = head->next;
        temp->next = head;
        delete cN;
        cout << "Student removed successfully." << endl;
        return;
    }

    pN = cN;
    cN = cN->next;

    while(cN != head) {
        if(cN->data.getID() == studentID) {
            pN->next = cN->next;
            delete cN;
            cout << "Student removed successfully." << endl;
            return;
        }
        pN = cN;
        cN = cN->next;
    }
    cout << "Student not found." << endl;
}

void display() {
    if(head == NULL) {
        cout << "No records found." << endl;
        return;
    }

    cout << "Student Records:" << endl;
    Node* temp2 = head;
    do {
        cout << "ID: " << temp2->data.getID() << ", Name: " << temp2->data.getName() << ", GPA: " <<
temp2->data.getGpa() << endl;
        temp2 = temp2->next;
    } while(temp2 != head);
}

void searchStudent(int studentID) {
    if(head == NULL) {
        cout << "Student not found." << endl;
        return;
    }
    Node* temp = head;
    do {
        if(temp->data.getID() == studentID) {
            cout << "Student found:" << endl;

```

```

        cout << "ID: " << temp->data.getID() << ", Name: " << temp->data.getName() << ", GPA: " <<
temp->data.getGpa() << endl;
        return;
    }
    temp = temp->next;
} while (temp != head);

cout << "Student not found." << endl;
}

float calculateAvgGpa() {
    if(head == NULL) {
        return 0.0;
    }
    float sum = 0.0;
    int count = 0;
    Node *temp = head;
    do {
        sum += temp->data.getGpa();
        count++;
        temp = temp->next;
    } while (temp != head);
    return sum / count;
}
};

int main() {
    LinkedList l1;
    int choice;
    int id;
    string name;
    float gpa;
    cout << "Welcome to the Student Registration System!" << endl;
    cout << "1. Add a student" << endl;
    cout << "2. Remove a student" << endl;
    cout << "3. Search for student" << endl;
    cout << "4. Display all students" << endl;
    cout << "5. Calculate average GPA" << endl;
    cout << "6. Exit" << endl;

    do {
        cout << "Enter your choice: ";
        cin >> choice;
        switch (choice) {
            case 1:
            {
                cout << "Enter student ID: ";
                cin >> id;

```

```

        cout << "Enter Student name: ";
        cin.ignore();
        getline(cin, name);
        cout << "Enter student GPA: ";
        cin >> gpa;
        Student newStudent(id, name, gpa);
        l1.addStudent(newStudent);
        break;
    }

    case 2:
        cout << "Enter student ID to remove: ";
        cin >> id;
        l1.removeStudent(id);
        break;
    case 3:
        cout << "Enter student ID to search: ";
        cin >> id;
        l1.searchStudent(id);
        break;
    case 4:
        l1.display();
        break;
    case 5:
        cout << "Average GPA: " << l1.calculateAvgGpa() << endl;
        break;
    case 6:
        cout << "Goodbye!" << endl;
        break;
    default:
        cout << "Invalid choice. Please try again." << endl;
        break;
    }
} while(choice != 6);

return 0;
}

```

Question 2:

```

#include<iostream>
using namespace std;
class Node{
// private:
public:
    int data;
    Node *next;

```

```

    Node(int val = 0) : data(val) , next(NULL) {}
};
class LinkedList{
// private:
public:
    Node *head1, *head2;

    LinkedList() { head1 = NULL; head2 = NULL;}

    void insert1(int val) {
        Node *newNode = new Node(val);
        if(head1 == NULL) {
            head1 = newNode;
            head1->next = head1;
        }
        Node *temp = head1;
        while(temp->next != head1) {
            temp = temp->next;
        }
        temp->next = newNode;
        newNode->next = head1;
    }

    void insert2(int val) {
        Node *newNode = new Node(val);
        if(head2 == NULL) {
            head2 = newNode;
            head2->next = head2;
        }
        Node *temp = head2;
        while(temp->next != head2) {
            temp = temp->next;
        }
        temp->next = newNode;
        newNode->next = head2;
    }

    void checkCommon() {
        int count = 0;

        Node *i = head1;
        do{
            Node *j = head2;
            do{
                if(i->data == j->data) {
                    cout<<"common values: "<<i->data<<endl;
                    count++;
                }
            } while(j != NULL);
            i = i->next;
        } while(i != NULL);
    }
};

```

```

        }
        j = j->next;
    } while(j != head2);
    i = i->next;
} while( i != head1);

// for (Node *i = head1; i->next != head1; i = i->next) {
//     for(Node *j = head2 ; j->next != head2; j = j->next) {
//         if(i->data == j->data) {
//             cout<<"common values: "<<i->data<<endl;
//             count++;
//         }
//     }
// }
// }
cout<<"total common elements are: "<<count<<endl;
}
};
int main() {
    LinkedList l1;
    l1.insert1(10);
    l1.insert1(20);
    l1.insert1(30);
    l1.insert1(40);
    l1.insert1(50);

    l1.insert2(30);
    l1.insert2(50);
    l1.insert2(70);
    l1.insert2(90);

    l1.checkCommon();
}

```

Question 3:

```

#include <iostream>
using namespace std;

```

```

class Node {
public:
    int data;
    Node* next;

    Node(int val) {
        data = val;
        next = NULL;
    }
}

```



```
};
```

```
class LinkedList {
```

```
private:
```

```
    Node* head;
```

```
public:
```

```
    LinkedList() {
```

```
        head = NULL;
```

```
    }
```

```
    void insert(int val) {
```

```
        Node* newNode = new Node(val);
```

```
        if (head == NULL) {
```

```
            head = newNode;
```

```
            head->next = head;
```

```
        } else {
```

```
            Node* temp = head;
```

```
            while (temp->next != head) {
```

```
                temp = temp->next;
```

```
            }
```

```
            temp->next = newNode;
```

```
            newNode->next = head;
```

```
        }
```

```
    }
```

```
    void removeNode(int target) {
```

```
        Node* cN = head;
```

```
        Node* pN = NULL;
```

```
        if(head->data == target) {
```

```
            if(head->next == head) {
```

```
                delete head;
```

```
                head = NULL;
```

```
                return;
```

```
            }
```

```
            Node *temp1 = head;
```

```
            while (temp1->next != head) {
```

```
                temp1 = temp1->next;
```

```
            }
```

```
            Node *temp2 = head;
```

```
            head = head->next;
```

```
            temp2->next = head;
```

```
            delete temp2;
```

```
            return;
```

```

    }
    do {
        pN = cN;
        cN = cN->next;
        if(cN->data == target) {
            pN->next = cN->next;
            delete cN;
            return;
        }
    } while (cN != head);
}

void find() {
    if(!head) return;
    Node* i = head;
    do {
        Node* j = i->next;
        Node* prev = i;
        while(j != head) {
            if (j->data == i->data) {
                prev->next = j->next;
                delete j;
                j = prev->next;
            } else {
                prev = j;
                j = j->next;
            }
        }
        i = i->next;
    } while ( i != head);
}

void display() {
    Node* temp = head;
    do {
        cout << temp->data << " ";
        temp = temp->next;
    }

    while (temp != head);
}

};

int main() {
    LinkedList l1;
    l1.insert(10);
    l1.insert(20);
    l1.insert(30);
    l1.insert(20);
    l1.insert(40);

```

```
l1.insert(30);

l1.display();
l1.find();
// l1.find();~
cout<<endl;
l1.display();
}
```